

WHEN BETTER GETS WORSE: IMPROVEMENT FIDELITY OF SELF-IMPROVEMENT OPERATORS IN ADAPTIVE WORLDS

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ABSTRACT

Self-improvement systems repeatedly replace an incumbent policy with a candidate that a verifier says is better. The hidden assumption is that this verifier-approved improvement transfers after deployment, even when the deployment changes the world. This implication can fail even when the verifier correctly evaluates every individual policy.

We therefore define *Improvement Fidelity* over the update transition $\tau = (\pi_t, \pi_{t+1})$, rather than over isolated policy values. The proxy improvement is $\Delta_V = J_V(\pi') - J_V(\pi)$ and the deployment-induced improvement is $\Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi)$. An *improvement reversal* is $\Delta_V > 0$ with $\Delta_* < 0$: a verifier-approved replacement that becomes worse after the world responds.

PIVOT (Paired Interventional Validation of Optimization Transitions) is a budgeted intervention mechanism that preserves update decisions. Its primary rule, PIVOT-VOI, maintains a posterior over the differential correction $\Delta_* - \Delta_V$ and queries the transition with largest expected value of sample information per cost; the earlier transparent score is retained as PIVOT-H. It is not generic active learning and does not target global world-model error.

Across the preregistered V9 ladder, E2C and E3C support the registered operator-shift and closed-loop contrasts, E4C is a powered null for the learned OOD evaluator contrast, and E5C and E7C support scoped efficiency and strategic response contrasts. The finance stress test remains observational. We do not claim causal market reversal, simulator ground truth, universal failure, or general multi-agent equilibrium; the contribution is a falsifiable evaluation criterion and a decision-theoretic implementation.

1 INTRODUCTION

Self-improvement is often described as a loop: generate a candidate, evaluate it, promote it, and repeat. But the central object optimized by the loop is not a policy in isolation; it is a *replacement operation*. A verifier may rank policies correctly while ranking improvements incorrectly, because deployment can change the transition and reward dynamics that define the next evaluation. The hidden implication is

$$\text{verifier says candidate is better} \implies \text{deployment policy is better}, \quad (1)$$

and this implication need not hold even for a fixed, external, correctly implemented verifier.

This failure is distinct from self-authored verification, ordinary dataset overfitting, or an incorrect evaluation program. It is a transfer failure for the learning update itself: the proxy world evaluates an object that changes when the candidate is deployed.

We study the local transition

$$\tau_t = (\pi_t, \pi_{t+1}) \quad \text{rather than only} \quad J(\pi_t). \quad (2)$$

Let V be a cheap observer or replay world and let $*$ denote a deployment world that may respond to the policy. The proxy and deployment improvements are $\Delta_V = J_V(\pi') - J_V(\pi)$ and $\Delta_* =$

$\mathcal{J}(\pi') - \mathcal{J}(\pi)$. An improvement reversal is

$$\Delta_V > 0 \quad \wedge \quad \Delta_* < 0. \quad (3)$$

This definition makes no assumption about whether the proxy is learned, hand-coded, or externally audited. It asks whether the object consumed by the self-improvement loop transfers.

An improvement operator $\mathcal{A}$ induces a transition distribution $Q_{\mathcal{A}}$ over the replacements it proposes. We therefore define operator-relative Improvement Fidelity for any loss L by

$$\text{IF}(V, \mathcal{A}; L) = \mathbb{E}_{(\pi, \pi') \sim Q_{\mathcal{A}}} [L(\Delta_V, \Delta_*)]. \quad (4)$$

The absolute-delta choice $L(a, b) = |a - b|$ is operator-conditioned IDE, while the sign-error choice is $1 - \text{ISC}(Q_{\mathcal{A}})$ after excluding ties. IRR is the corresponding conditional failure probability. Thus the relevant population is induced by the self-improvement operator, rather than the uniform population of all policy pairs.

Our contributions are deliberately limited to three points:

1. **Contribution 1 — Principle and theory.** We introduce operator-relative *Improvement Fidelity*, prove Global Fidelity Blindness and an Operator Shift Bound, and retain the local response-footprint bound.
2. **Contribution 2 — Algorithm.** We introduce **PIVOT-VOI**, a paired posterior differential evaluator with EVSI-per-cost acquisition, adaptive stopping, and an explicit best-update sample bound; PIVOT-H is a transparent heuristic baseline.
3. **Contribution 3 — Evidence contract.** We provide controlled operator-shift and response-mechanism evidence plus a pinned external adaptive/strategic pipeline. Development underpowering and the negative finance boundary are reported as scientific outcomes, not hidden.

2 RELATED WORK

Performative and strategic response. Performative prediction makes deployment part of the data-generating process Perdomo et al. (2020); Mandal et al. (2022), and recent work derives policy gradients or learning dynamics for performative RL and performative Markov games Basu et al. (2025); Sahitaj et al. (2025); Wang et al. (2025); Piliouras & Yu (2022). These papers study stable policies, equilibria, or convergence. They do not ask whether a particular replacement $\pi \rightarrow \pi'$ is still an improvement after the response is realized. **Gap:** PIVOT treats the update, not the fixed point, as the estimand.

World-model and simulator fidelity. WorldGym and policy-aware simulator learning evaluate policy values, rankings, or simulator objectives Quevedo et al. (2025); Dann et al. (2026); Hansen et al. (2024). Recent market simulators improve the realism of endogenous order flow and liquidity through interactive LOB feedback or generative state-event models Noble et al. (2026); Zhang et al. (2026); Kawawa-Beaudan et al. (2026). These advances still report policy-level fidelity or trajectory realism. **Gap:** they do not measure the sign or magnitude of the differential $J(\pi') - J(\pi)$ under a policy-induced world, so good global ranking can coexist with wrong local update ordering.

Self-improvement and verification. EVOQUANT, EvoPolicyGym, Darwin Godel Machine, and Evo-Harness make iterative policy or harness evolution executable Mao et al. (2026); Wang et al. (2026); Zhang et al. (2025); Wei et al. (2026); FinEvo-Bench extends evaluation to longitudinal financial workflows Deng et al. (2026). ICLR benchmarks such as AgentBench, WebArena, and SWE-bench make agent evaluation executable across interactive tasks Liu et al. (2024); Zhou et al. (2024); Jimenez et al. (2024). SEAL identifies the verifier-deployment gap when an agent authors its own tests, while recent stress tests expose variance, task-order, and execution-interface failures Guo et al. (2026); Ye et al. (2026); Li & Zhu (2026); Kulkarni et al. (2026). Those works either protect acceptance with a sealed oracle or benchmark the resulting agent. **Gap:** even a fixed external verifier can be correct about the wrong world; PIVOT validates transfer of the proposed update itself.

Decision-preserving queries and finance testbeds. Active learning and value-of-information provide the ancestry for allocating expensive queries Settles (2009); Lindley (1956); paired rollouts additionally exploit shared exogenous randomness. PIVOT changes the target from global prediction

error to update-selection regret. ABIDES-MARL, FinEvo, and the NeurIPS FinRL-Meta benchmark, together with recent execution and market-impact simulators, supply increasingly endogenous finance testbeds Cheridito et al. (2025); Zou et al. (2026); Liu et al. (2022); Byrd et al. (2019); Cont et al. (2010); Nevmyvaka et al. (2006). **Gap:** these environments can reveal response mechanisms, but none defines improvement consistency or a budgeted transition-level validator that separates observer, actor, and strategic reversals.

3 PROBLEM FORMULATION

Policy-induced worlds. Deployment can alter transitions, rewards, liquidity, or other agents' behavior. We write this response as $\mathcal{M}[\pi]$ and define the deployment value

$$\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi]). \quad (5)$$

For exposition, three layers separate mechanisms rather than assert that they are always identifiable:

$$\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \quad (6)$$

$$\Delta_{\text{actor}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \quad (7)$$

$$\Delta_{\text{strategic}} = J_i(\pi', BR_{-i}(\pi')) - J_i(\pi, BR_{-i}(\pi)). \quad (8)$$

The first ignores response, the second includes the focal agent's mechanical footprint, and the third includes adaptation by competitors. In experiments we report these as separate worlds and do not label a learned or observational proxy as ground truth.

Improvement fidelity metrics. For n transitions, IDE is $n^{-1} \sum_k |(\Delta_V)_k - (\Delta_*)_k|$. ISC is the fraction of non-tied transitions whose signs agree. IRR is $P(\Delta_* < 0 \mid \Delta_V > 0)$, while SIRR is $P(\Delta_{\text{strategic}} < 0 \mid \Delta_{\text{actor}} > 0)$. When the rows are draws from $Q_{\mathcal{A}}$, IDE and $1 - \text{ISC}$ are the two empirical losses in $\text{IF}(V, \mathcal{A}; L)$ above; the operator label is therefore part of the estimand, not merely metadata. For a candidate set C_t , the update-selection regret is

$$\text{ISR}_t = \max_{j \in C_t} (\Delta_*)_{t,j} - (\Delta_*)_{t,\hat{j}}, \quad (9)$$

where $\hat{j}$ is selected by the proxy or corrected estimate. CTI sums the true improvements of promoted updates over a trajectory. We call this metric the cumulative true improvement (CTI). All estimates below retain query counts, seeds, and confidence intervals in the artifact snapshot.

4 PIVOT

PIVOT stands for Paired Interventional Validation of Optimization Transitions. Given an incumbent and K candidates, it first computes cheap proxy deltas and pre-query features. The features include proxy delta, response strength, competition strength, candidate index, and an update footprint. The generic footprint records policy-distance and action/support shifts; the finance footprint records size, participation, urgency, holding, turnover, liquidity consumption, and spread crossing.

Paired differential correction. For a queried transition, incumbent and candidate are evaluated on the same initial state, exogenous path, seed, and opponent initialization. PIVOT fits

$$g_\theta(z_\Delta) \approx \Delta_* - \Delta_V, \quad \hat{\Delta}_* = \Delta_V + g_\theta(z_\Delta). \quad (10)$$

Pairing subtracts common-mode noise before aggregation. The unpaired ablation uses disjoint context seeds and the standard independent-sample standard error.

Active query rule. Let D denote current paired evidence and let the posterior over true deltas define expected simple regret $\mathcal{R}(D) = \mathbb{E}[\max_j \Delta_j^* - \Delta_{a(D)}^* \mid D]$, where $a(D) = \arg \max_j \mathbb{E}[\Delta_j^* \mid D]$. A hypothetical HF observation Y_j induces

$$\text{EVSI}_j = \mathcal{R}(D) - \mathbb{E}_{Y_j \mid D}[\mathcal{R}(D \cup \{Y_j\})], \quad A_j = \frac{\text{EVSI}_j}{c_j}, \quad (11)$$

and PIVOT-VOI queries $\arg \max_j A_j$. It stops when the posterior selection probability is at least $1 - \delta$ or $\max_j A_j < \eta$. The earlier transparent score $q_j = u_j(0.5 + r_j)(1 + o_j)/c_j$ is PIVOT-H, a baseline rather than the primary claim. Queried values replace predictions; unqueried candidates retain proxy/model estimates.

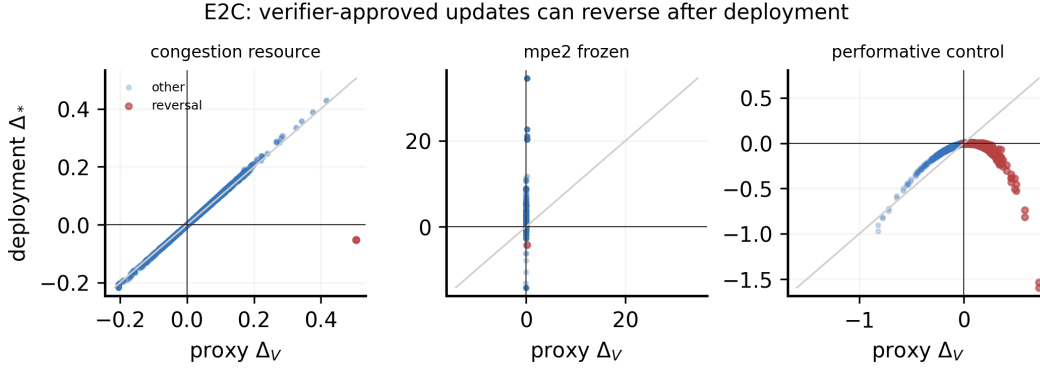


Figure 1: E2C paired proxy versus deployment improvements, split by environment. Red points are improvement reversals. The panel is descriptive evidence for the registered response mechanisms, not market ground truth.

Why Transition Validation Differs from Active Learning. Traditional active learning asks which label will reduce a global prediction loss for y . PIVOT asks which paired intervention can change the ordering or sign of an update. Its target is the differential correction $\Delta_* - \Delta_V$, and its loss is update-selection regret rather than global world-model error. In short, **PIVOT is not designed to minimize global world-model error; it targets the smallest intervention budget required to preserve optimization decisions.** This distinction is operational: an uncertain transition whose high-fidelity value cannot change the selected candidate has lower priority than an otherwise similar transition near the selection margin.

5 THEORY

We state four results that motivate the differential estimand. They are not claims about a particular simulator architecture.

Proposition 1 (Global fidelity is sufficient). *Suppose $\sup_\pi |J_V(\pi) - J_*(\pi)| \leq \varepsilon$. Then every transition satisfies $|\Delta_V - \Delta_*| \leq 2\varepsilon$. In particular, the sign is preserved whenever $|\Delta_*| > 2\varepsilon$.*

Proof. By adding and subtracting the two policy values, $\Delta_V - \Delta_* = [J_V(\pi') - J_*(\pi')] - [J_V(\pi) - J_*(\pi)]$. The triangle inequality gives the bound. If $|\Delta_*| > 2\varepsilon$, the perturbation cannot cross zero. $\square$

Proposition 2 (Global Fidelity Blindness). *For every $\varepsilon > 0$, there is a finite policy family, a deployment value function J_* , a verifier V , a policy distribution P , and an update operator $\mathcal{A}$ such that*

$$\text{MAE}_P(J_V, J_*) < \varepsilon, \quad \text{Spearman}(J_V, J_*) > 1 - \varepsilon,$$

while, for the transition law $Q_{\mathcal{A}}$,

$$\text{IRR}(Q_{\mathcal{A}}) = 1, \quad \text{ISC}(Q_{\mathcal{A}}) = 0.$$

Proof. Take n policies with uniform P and $J_*(i) = i/(n-1)$, and let V swap the values of one adjacent pair. Then $\text{MAE}_P = 2/[n(n-1)]$ and the only nonzero rank displacements are $+1$ and -1 , so $1 - \text{Spearman} = 12/[n(n^2-1)]$. Choose n large enough that both quantities are below ε . Let $\mathcal{A}$ sample only the swapped directed transition (so $Q_{\mathcal{A}}$ has unit mass there) and propose the lower-valued true policy as a replacement for the higher-valued incumbent. The proxy delta is positive and the deployment delta is negative on every sampled transition, giving $\text{IRR}(Q_{\mathcal{A}}) = 1$ and $\text{ISC}(Q_{\mathcal{A}}) = 0$. $\square$

Proposition 3 (Operator Shift Bound). *Let P be the calibration law over transitions and let $Q_{\mathcal{A}} \ll P$ have density ratio $w = dQ_{\mathcal{A}}/dP$. For a non-negative transition loss $\ell(\tau) = L(\Delta_V(\tau), \Delta_*(\tau))$,*

$$\text{IF}(V, \mathcal{A}; L) = \mathbb{E}_P[w(\tau)\ell(\tau)] \leq \sqrt{\mathbb{E}_P[\ell(\tau)^2]} \sqrt{1 + \chi^2(Q_{\mathcal{A}}\|P)}. \quad (12)$$

Proof. Absolute continuity gives the first equality. Cauchy–Schwarz bounds the expectation by $\sqrt{\mathbb{E}_P[\ell^2]} \sqrt{\mathbb{E}_P[w^2]}$. Since $\mathbb{E}_P[w^2] = 1 + \chi^2(Q_A \| P)$, the result follows. $\square$

Proposition 4 (Response–footprint bound). *Let $d(\pi, \pi')$ be an update footprint and define the direct comparison in the incumbent response world, $\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi])$. Assume the environment response satisfies $D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \leq L_M d(\pi, \pi')$ and the value functional is L_J -Lipschitz in its world argument. Then*

$$|\Delta_{\text{actor}} - \Delta_{\text{direct}}| \leq L_J L_M d(\pi, \pi'). \quad (13)$$

Proof. The incumbent term cancels because both comparisons use $\mathcal{M}[\pi]$ as the reference world. The remaining candidate term is bounded by $L_J D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \leq L_J L_M d(\pi, \pi')$. The result is local; it does not assert that strategic best responses are globally Lipschitz. $\square$

Proposition 5 (Decision Preservation Under Differential Error). *Consider two candidates i and j evaluated from the same incumbent, and suppose the true improvement margin is $m = \Delta_{*,i} - \Delta_{*,j} > 0$. If estimates satisfy $|\hat{\Delta}_i - \Delta_{*,i}| < m/2$ and $|\hat{\Delta}_j - \Delta_{*,j}| < m/2$, then $\hat{\Delta}_i > \hat{\Delta}_j$ and the selected update is unchanged.*

Proof. The estimated margin obeys $\hat{\Delta}_i - \hat{\Delta}_j = m + (\hat{\Delta}_i - \Delta_{*,i}) - (\hat{\Delta}_j - \Delta_{*,j})$. The two error terms have absolute sum strictly less than m , so the estimated margin remains positive. This is a local decision guarantee; it does not claim that PIVOT is globally optimal or that all candidates satisfy the margin condition. $\square$

Proposition 6 (Finite-Sample Best-Update Identification). *Suppose paired rollout differences are σ -sub-Gaussian and the best candidate has true margin at least $m > 0$ over each of $K - 1$ competitors. With n paired rollouts per candidate, the probability of selecting a wrong update is at most*

$$(K - 1) \exp\left(-\frac{nm^2}{4\sigma^2}\right). \quad (14)$$

Thus $n \geq 4\sigma^2 m^{-2} \log((K - 1)/\delta)$ suffices for error at most δ (and the implementation uses the conservative K in its registry).

Proof. For each competitor, a wrong ordering requires the difference of two sample-mean errors to exceed m . The sub-Gaussian tail bound gives the stated pairwise probability; a union bound over $K - 1$ competitors completes the proof. $\square$

The first proposition explains why global value accuracy is sufficient but unnecessarily strong: a policy-independent offset can be arbitrarily large without changing any improvement difference. Global Fidelity Blindness shows that the converse fails when the update law concentrates on a locally swapped transition. The Operator Shift Bound quantifies the bridge: local IF can grow with the chi-square distance between the calibration law and Q_A . The response–footprint result motivates targeted fidelity, while the final proposition connects differential error to the selection decision PIVOT must preserve. In strategic environments, small policy changes can still cause large competitor responses, so smoothness remains a diagnostic assumption.

6 CONFIRMATORY EXPERIMENTS

Design. The preregistered V9 suite uses two newly implemented environment families (performative and congestion/resource), three candidate operators, ten shift levels, paired common-random-number rollouts, and 30 independent seeds. The powered Farama MPE2 result is retained as a frozen external null and joined only as a separate reference panel; it is not silently rerun or relabeled. The unit of inference is the seed or trajectory cluster; transition rows are retained for audit but are not treated as independent observations.

The resulting decision-preserving data flow is summarized in fig. 3. The diagram treats a policy replacement as the unit of validation: cheap proxy scores screen the candidate set, while paired high-fidelity interventions are reserved for transitions that can change the selection decision.

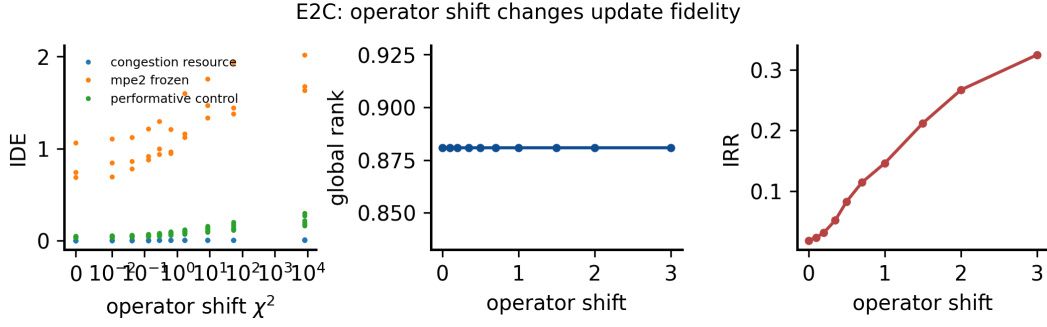


Figure 2: E2C operator-shift scaling. IDE rises with the registered shift while global rank remains comparatively stable; IRR is the transition-level failure probability.

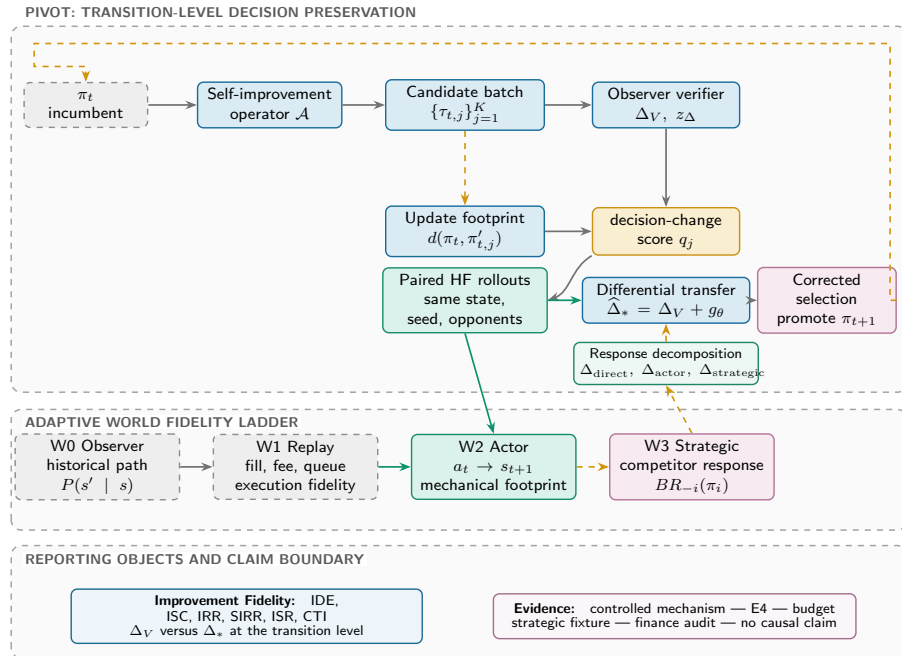


Figure 3: PIVOT validates the replacement operation rather than isolated policy values. A cheap verifier produces proxy improvements and update footprints; a decision-change score selects paired high-fidelity interventions under common state, seed, and opponent conditions. The differential correction ranks candidates for promotion and feeds the next self-improvement round.

Value Fidelity versus Improvement Fidelity. Global policy-value rank and transition-level update fidelity are related but not interchangeable. The E2C raw sampled reversal-rate cells preserve both quantities: a global evaluator can remain competitive while the differential sign fails on the operator-induced population. We use “false improvement” and “improvement reversal” for the same event, $\Delta_V > 0$ and $\Delta_* < 0$; this is the **false improvement (improvement reversal)** event.

E2C: operator-shift mechanism. The confirmatory E2C run has 30 seeds and 36000 paired transition rows. The high-minus-low shift effect on operator-relative IDE is 0.2627 with bootstrap interval [0.1684, 0.3763], yielding status HYPOTHESIS.SUPPORTED. Global rank remains comparatively stable in the same cells, so this is evidence for the update-conditioned estimand rather than an assertion that global policy ranking fails.

E3C–E5C: closed loop, OOD, and budget. E3C runs 40 rounds with eight candidates and ten methods. PIVOT-VOI versus Proxy Only has paired CISR reduction 1.4883 with interval

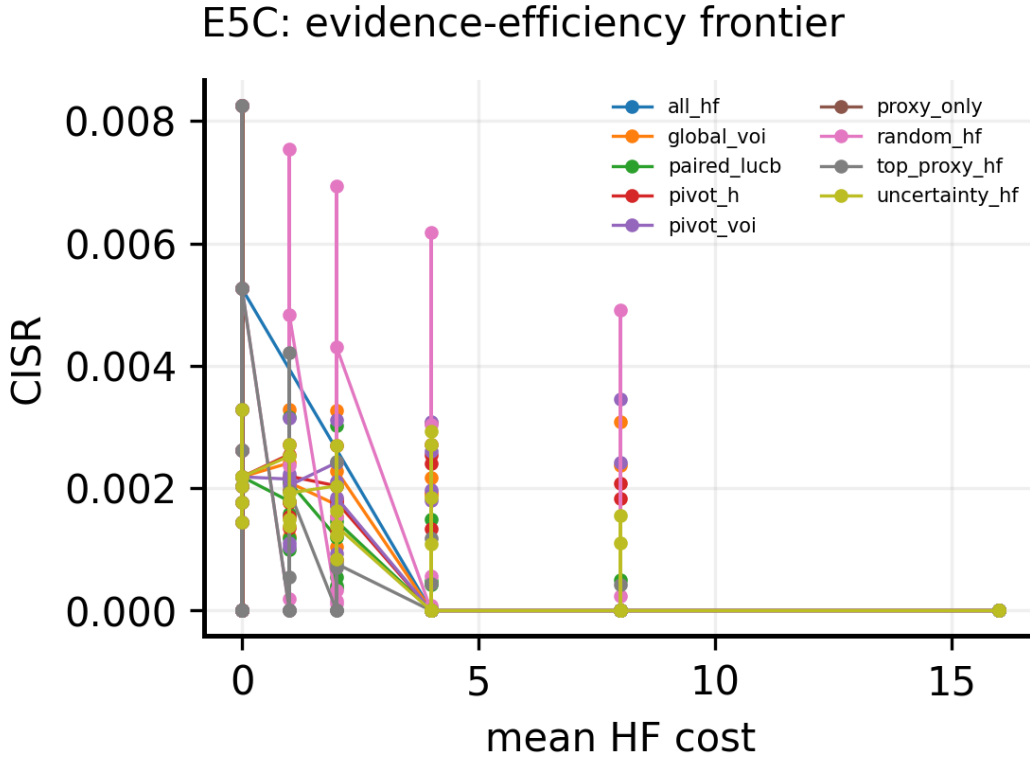


Figure 4: E5C fixed-budget frontier across candidate counts and environments. All-HF is shown only as an oracle reference and is not a comparable acquisition method.

[1.0618, 1.9105] (HYPOTHESIS_SUPPORTED); its comparison with Global-VOI and LUCB is not uniformly favorable. The E4C learned-evaluator comparison is powered but negative: differential-minus-global transition ISC is -0.2590 with interval [-0.3588, -0.1707], so no broad OOD transfer claim is made. At fixed E5C high-fidelity budget, PIVOT-VOI’s paired CISR reduction over Proxy Only is 0.000668 with interval [0.000011, 0.0013] (HYPOTHESIS_SUPPORTED).

7 STRESS TESTS BEYOND CONTROLLED ENVIRONMENTS

Pinned external and strategic boundary. The frozen MPE2 reference remains an explicit external null and is not relabeled by V9. E7C uses only a focal self-improving policy and independent opponent mechanisms. Across best-response, gradient-adaptive, and RL/evolutionary modes, the cluster strategic effect is -0.0240 with interval [-0.0249, -0.0231] and mean SIRR 0.9495 (HYPOTHESIS_SUPPORTED). This is scoped to the registered opponent families; it is not an equilibrium, ecology, or market-ground-truth claim.

Strategic response. Only the focal policy self-improves. The fixed opponent remains near zero effect, while independent adaptive mechanisms introduce a negative strategic layer. Diagnostics are in Appendix figs. 6 and 9.

Finance audit. The finance ladder is F0 historical path, F1 execution replay, F2 virtual fills, and F4 strategic response. Across 12 BTCUSDT, ETHUSDT, and BNBUSDT sessions, the public-data depth proxy has reversal 0/7 (primary) and 0/5 (holdout), with pooled effect -4.1554×10^{-7} . It is a plausibility/boundary test, not causal confirmation.

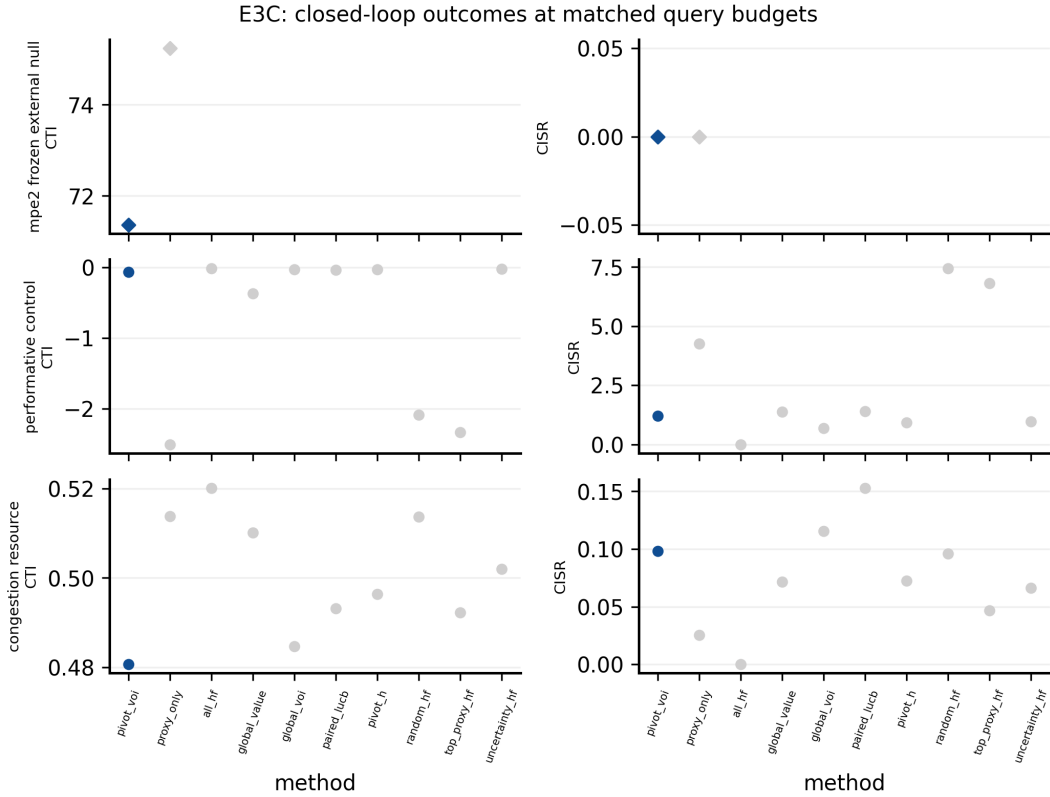


Figure 5: E3C closed-loop CTI and CISR at matched query budgets. The top row is the frozen powered MPE2 external null (diamonds; V7 source), while the lower rows are the two newly implemented V9 environments. PIVOT-VOI is highlighted; the full method matrix remains in the source table and appendix.

Table 1: Historical V7 ablation boundary. V9 confirmatory efficiency results are reported in fig. 4 and the generated V9 tables.

Ablation	Estimate(s)	Metric
Paired vs unpaired	0.0433 / 0.0676	paired SE / unpaired SE
Footprint vs none	2.0668 / 2.0669	IDE; null
Small vs large	0.50 / 0.75	IRR
Weak vs strong response	0.458 / 1.000	IRR
Fixed vs adaptive	0.000 / 1.000	SIRR
Single vs multiple model	4.751 / 2.691	IDE
Candidate count 2 / 4	0.2368 / 0.2439	ISR
PIVOT budget 0 / 4	1.0210 / 0.0000	ISR

8 ABLATIONS AND ROBUSTNESS

The V9 evidence-efficiency ablation spans candidate counts $K \in \{4, 8, 16\}$, zero-to-all high-fidelity budgets, nine acquisition rules, posterior sample grids, and cost misspecification. The full frontier, query-set stability, and oracle flag are generated from the frozen V9 source tables. The historical V7 ablation table is retained in the supplement as a boundary artifact rather than as a V9 confirmatory estimate.

The V9 efficiency contrast is scoped to the registered candidate counts, budgets, and environments. Query-set stability varies across posterior sample and cost grids, and PIVOT-VOI is not uniformly best against every baseline.

9 LIMITATIONS AND CLAIM BOUNDARY

The controlled worlds establish falsifiability rather than realism. E4C is a powered null for the registered learned OOD contrast; E3C is positive against Proxy Only but not uniformly against Global-VOI or LUCB. Finance is observational with virtual fills and no live or private data. E7C is bounded to the registered opponent mechanisms. LLM/EvoQuant and M3 adapters remain deferred. We do not certify actions, solve equilibrium, or call a learned simulator ground truth.

10 REPRODUCIBILITY AND CONCLUSION

The release contains the immutable V7 pre-upgrade snapshot, five V9 experiment registries, seven vector figure/source bundles, Parquet tables, per-run provenance and manifests, and automated statistical and claim audits. The confirmatory runs use fixed configurations and 30 independent seeds; no credential, raw vendor archive, live order, or private path enters the package.

The lesson is that self-improvement must be judged by whether an update remains better in the world—and among the agents—that deployment causes to exist. PIVOT implements paired, transition-level, budgeted interventional validation; E2C supports the operator-shift phenomenon, E3C and E5C support scoped decision-preservation contrasts, E4C is a powered null, and E7C exposes a registered strategic response layer. These are evidence about the estimand and testbed mechanisms, not universal claims about self-improving agents.

AI Use Statement. AI tools provided limited assistance with language editing and code/figure formatting; the authors verified all claims, computations, and artifacts and take full responsibility.

Reproducibility Statement. The anonymous supplement contains source, deterministic configurations, hash-indexed snapshot, table/figure inputs, and rebuild commands. The public-data audit uses virtual fills only and includes no credentials, private data, raw vendor archives, or live orders.

Ethics Statement. Public market data and controlled virtual environments only; no human subjects, personal data, live trading, or external authorization.

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A ARTIFACT AND FULL RESULT NOTES

The frozen V7 artifact manifest is `../snapshot/manifest.json`; the V9 run manifests and generated source tables live under `../results/v9`, `../figures/v9`, and `../tables/v9`. Re-running the V9 builders from the registered configurations is required before changing any result table or figure.

Table 2: Controlled and public evidence summary.

Experiment	Estimate	Interpretation
P2 reversal	0.9722	IRR, high response
E4 local-global	0.1204	ISC difference
E5 Random-PIVOT	4.7245	ISR at HF budget 1
E5 Top Proxy-PIVOT	3.1301	ISR at HF budget 1
E6 zero participation	0.0000	F2-F1
E7 SIRR	1.0000	strategic reversal
E8 SIRR	1.0000	strategic reversal
E9 CTI	-1.5338	8-round exploratory loop

Table 3: Public expansion boundary.

Public audit field	Value
Asset/date sessions	12
F1-positive sessions	7
Primary reversals	0/7
Holdout reversals	0/5
Pooled depth effect	-4.1554×10^{-7}
Causal impact identified	No

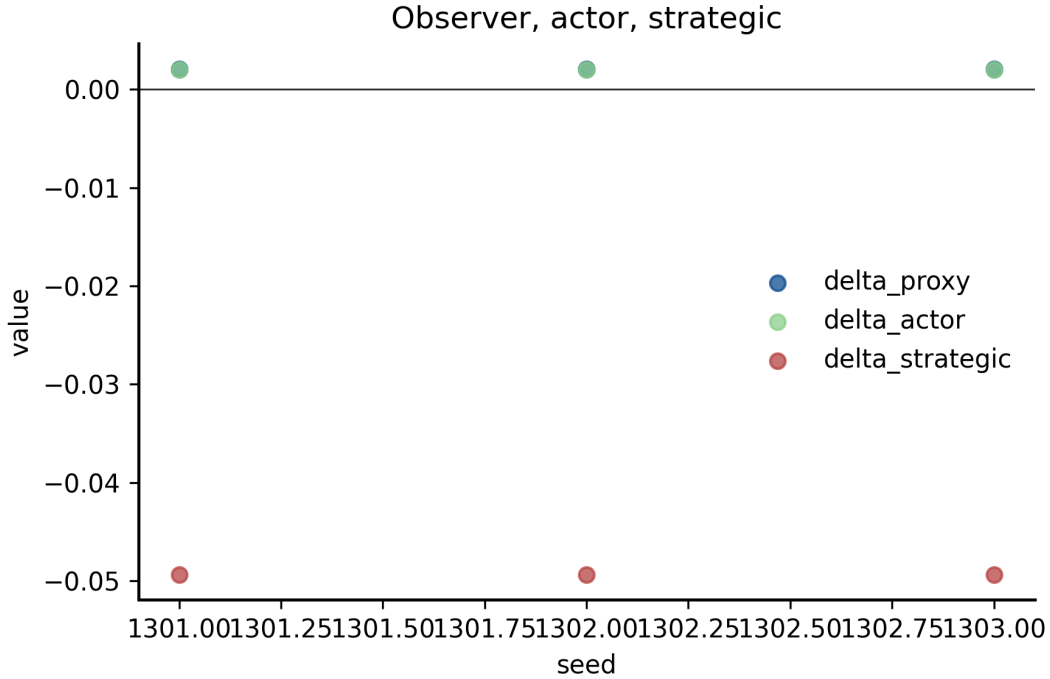


Figure 6: The same update evaluated in observer, actor, and strategic worlds. This diagnostic separates mechanical response from competitor adaptation; it is fixture evidence rather than a general-equilibrium claim.

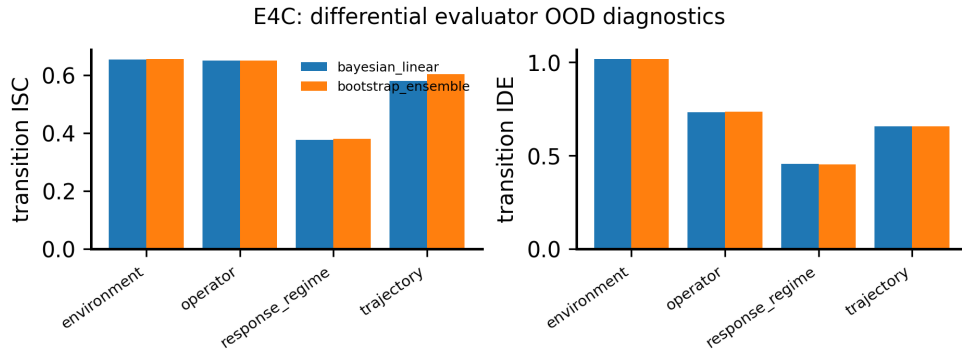


Figure 7: Powered E4C OOD diagnostic. The registered differential learner does not improve transition ISC over the global learner in this suite.

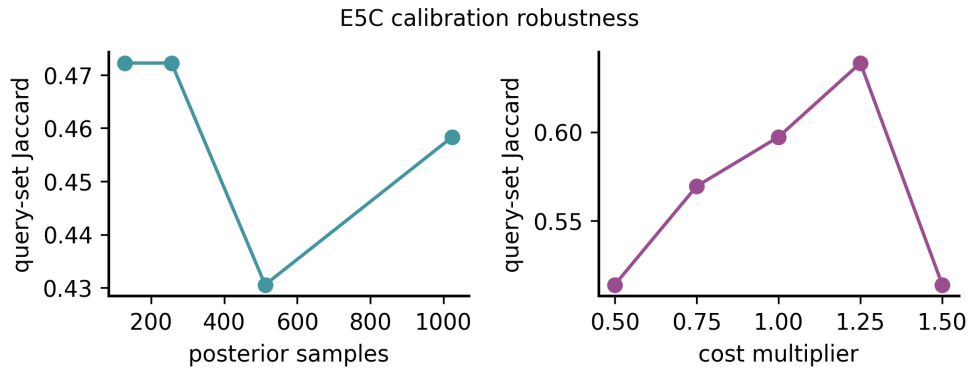


Figure 8: E5C posterior sample and cost-misspecification stability. These are calibration diagnostics, not additional superiority claims.

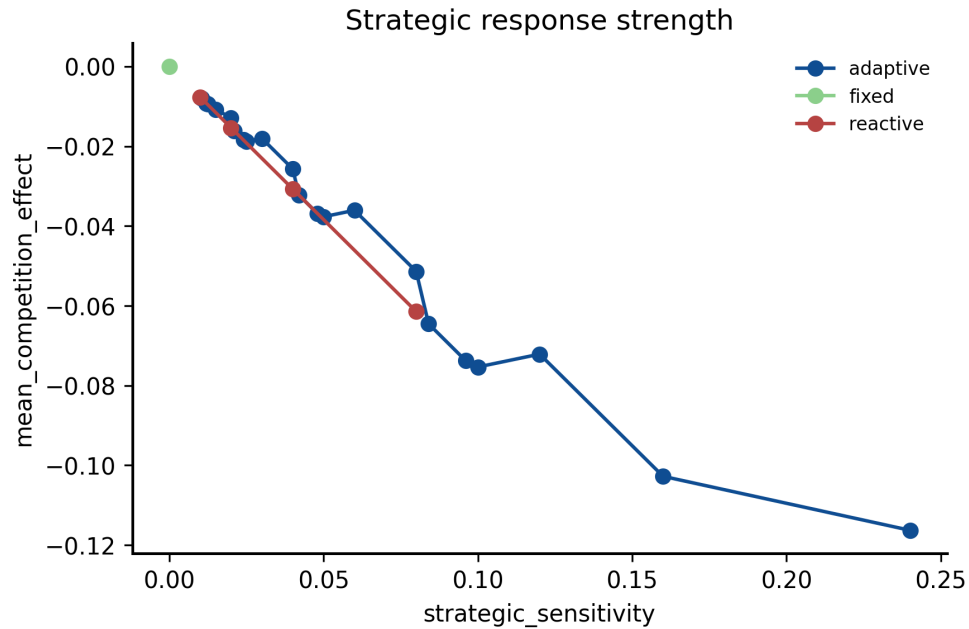


Figure 9: Strategic response effect across opponent modes and sensitivities. This appendix diagnostic is fixture evidence, not a general-equilibrium claim.